Debate Does Not Repair Encoding-Fragile Graph Reasoning: A Negative Result Where Verification Should Have Been Easy

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Abstract

Large language models read graphs only after the graph has been serialized into text, and accuracy swings sharply with the serialization. Multi-agent debate is a natural remedy: its premise is a *verification asymmetry*, the assumption that checking a claim is cheaper than producing one. That premise is normally asserted rather than measured, because in the domains where debate is usually tested the intermediate steps are not objectively checkable. Graph reasoning removes that excuse: an edge claim is either in the encoding or it is not. We build a Proposer–Critic debate on GraphQA (3 tasks $\times$ 3 encodings $\times$ 600 graphs, Qwen2.5-3B-Instruct) and measure the asymmetry directly. It is absent. Pooled over 6,424 verdicts the Critic's REVISE is statistically independent of whether the answer it judged was wrong ($\phi = +0.02$, $p = 0.07$), and 4–26% of the edges it cites as evidence do not exist in the graph. Debate leaves mean accuracy unchanged (0.496 vs. 0.501) but is far from inert per cell, moving accuracy by -0.100 to $+0.095$; it more than doubles the encoding spread on CONNECTED_NODES and manufactures a significant encoding effect on EDGE_EXISTENCE, a task with none at baseline. We localize the cause to the chain-of-thought scaffold rather than to the debate loop, and bound the loop's remaining headroom at $+0.062$ with an oracle stopping rule.

1 Introduction

A graph must be serialized before a language model can read it, and how it is serialized matters. Fatemi et al. (2024) show that zero-shot accuracy on elementary graph questions varies sharply with the encoding, the task, and the graph, with no encoding dominating. This is a practical liability wherever relational data meets an LLM: the same model, the same graph, and a different wording give a different answer.

Multi-agent debate (Du et al., 2023) is an appealing remedy, and its appeal rests on an assumption. Irving et al. (2018) motivate debate by a *verification asymmetry*: checking a claim should be cheaper than producing one, so a critic of equal capacity can police a proposer. In the domains where debate is usually evaluated — math word problems, open-ended QA — that assumption cannot be tested, because the intermediate steps are not objectively checkable. It is therefore unclear whether the existing null results for debate (Huang et al., 2024; Choi et al., 2025) show that debate does not work, or only that verification happened to be hard there.

Graph reasoning removes the ambiguity. An atomic claim is "edge (u, v) is present," and the encoding settles it by string lookup. If the verification asymmetry exists anywhere, it exists here. We therefore ask:

> Does a Proposer–Critic debate make graph reasoning robust to how the graph is written, and does any gain come from the debate procedure rather than from extra compute?

The answer is no on both counts, and the interesting part is why. Our contributions:

1. We reproduce encoding fragility over 5,400 instances with paired tests, and identify EDGE_EXISTENCE as a non-fragile control (§4.1).
2. We show self-consistency at $10\times$ compute is indistinguishable from greedy in all nine cells, with a structural reason: terse answers leave no diverse reasoning paths to marginalize (§4.2).
3. We measure the assumed verification asymmetry and find it absent: the verdict carries $\approx$1 accuracy point of information, and in two cells it is *anti*-correlated with correctness (§4.5).

4. We show debate's effect is not null but *bidirectional* — it redistributes accuracy across encodings, amplifying the fragility it targeted — and attribute that to the chain-of-thought scaffold, not the loop (§4.3, §4.4).

2 Related Work

Graph encodings. Fatemi et al. (2024) introduce GraphQA and establish that LLM graph reasoning is encoding-dependent; on listing a node's neighbors, accuracy moves from 19.8% to 53.8% by wording alone. GraphToken (Perozzi et al., 2024) learns a soft encoding and raises absolute performance, but the task-dependent sensitivity remains. Both lines characterize or shift fragility; neither offers a mechanism that removes it at inference time, which is the gap we probe.

Debate and its nulls. Du et al. (2023) report gains from multi-agent debate; the gains are contested. Huang et al. (2024) show LLMs cannot reliably self-correct without external feedback, and Choi et al. (2025) attribute most reported debate gains to the majority vote implicit in aggregation rather than to the exchange of arguments, against the self-consistency baseline of Wang et al. (2023). Our contribution is not another null but a *measurement of the mechanism* those nulls leave unexplained: we instrument the critic's verdict against ground truth where verification is trivially checkable.

Chain of thought. Wei et al. (2022) show that eliciting intermediate steps improves reasoning. Any debate protocol inherits this, since a proposer emitting a verifiable trace is also doing CoT; prior evaluations rarely separate the two. We do, and the separation carries the effect.

3 Experimental Setup

Data. We use the released GraphQA generators and encoders (Fatemi et al., 2024) directly, so graph generation and encoder wording are byte-exact to the original. Graphs are Erdős–Rényi with $n \sim \mathcal{U}\{5, \ldots, 19\}$ nodes and $p \sim \mathcal{U}[0, 1]$; ground truth comes from NetworkX (Hagberg et al., 2008). Structure is held fixed at ER because our variable is encoding. We freeze three independent datasets of 200 graphs (seeds 7, 11, 13) and pool them, giving $n = 600$ per cell; because the three encodings are applied to the *same* graphs, all comparisons are paired.

Tasks and encodings. Three GraphQA tasks — EDGE_EXISTENCE (is (u, v) an edge?), NODE_DEGREE (degree of u), CONNECTED_NODES (list u's neighbors) — crossed with three encodings chosen to differ on *both* axes of node naming and edge phrasing: **adjacency** (integer nodes, parenthesized pairs), **incident** (integer nodes, natural-language neighbor lists), and **friendship** (named nodes, relational sentences). GraphQA's remaining encodings are cover-story variants of friendship. This gives nine task $\times$ encoding cells.

Model. Qwen2.5-3B-Instruct (Qwen Team, 2024) in fp16, one model serving both roles. The 3B size is forced by the 11 GB GPUs available to us; this is a deviation from the 7–8B models we planned, and we return to it in Limitations.

Conditions. **Baseline**: one greedy zero-shot answer, GraphQA's question verbatim plus a terse-format instruction, no CoT and no exemplars — faithful to the zero-shot row that produces the fragility result. **Majority vote**: $N = 10$ samples of the same prompt at $T = 0.7$ (top-p 0.8, top-k 20), resolved by the mode of the parsed draws. **Debate**: the Proposer answers with a numbered list of atomic edge claims and a final `ANSWER:` line; the Critic is instructed to re-derive the answer from the edge list and return `VERDICT: AGREE` or `VERDICT: REVISE` with each problem quoted from the graph; the Proposer then revises. The loop runs to consensus, to no-progress (a repeated answer), or to a budget of 10 responses, and the final answer is the last Proposer answer. Debate is greedy throughout, so it shares the baseline's decoding.

Compute accounting. We report cost per instance as *model responses* and *total tokens* (prompt + generated); generated-only would mislead, since a baseline instance averages 374 total tokens but under 5 generated — the encoded graph is the whole cost. Baseline spends 1 response / 374 tokens, majority vote 10 / 3,659 (10.0$\times$), debate 2.9 / 1,805 (4.8$\times$): debate is strictly *cheaper* than the vote it must beat.

Scoring and statistics. Accuracy is whole-answer exact match against NetworkX, with set equality for CONNECTED_NODES and no partial credit, mirroring GraphQA; a `parse_ok` flag keeps parse failures visible rather than silently scored wrong. Because encodings and conditions share graphs, significance is paired throughout: McNemar for pairwise deltas (exact binomial below 25 discordant pairs) and Cochran's Q as the omnibus test. In total we score 5,400 baseline instances, 5,400 debate instances (13,926 responses), and 18,000 majority-vote samples.

4 Results

4.1 Encoding fragility reproduces

The premise holds (Table 1, baseline column; Table 2, upper block). NODE_DEGREE spans 0.388 (adjacency) to 0.750 (incident), a gap of 0.362 with $Q = 200.2$, $p = 3\text{e}{-44}$; the best-versus-worst McNemar is 258/41, $p = 8\text{e}{-36}$. CONNECTED_NODES spans 0.217 to 0.343 ($Q = 65.8$, $p = 5\text{e}{-15}$). The direction replicates Fatemi et al. (2024): incident is best on both tasks.

EDGE_EXISTENCE is *not* encoding-fragile: spread 0.013, $Q = 0.52$, $p = 0.77$, and the best/worst labels differ across seeds — the signature of no effect. We treat it as a control, since an intervention has almost no fragility to remove there. It is also the task the Critic's atomic check reduces to.

4.2 Extra compute alone buys nothing

Majority vote at $N = 10$ is indistinguishable from a single greedy answer in all nine cells (seed 7, $n = 200$): every $|\Delta| \leq 0.010$, only 1–8 discordant instances per cell, and every McNemar $p \geq 0.625$. Voting returns greedy's exact correctness on 96–99% of instances, and the fragility gaps are unchanged.

The reason is structural. Self-consistency draws its leverage from marginalizing over *diverse reasoning paths* (Wang et al., 2023); our baseline emits a direct terse answer, so there are no paths to marginalize, and a vote over samples of a near-single-token answer converges to the argmax — exactly greedy decoding. This is the control debate needs: on these tasks 10× the compute spent on more samples of the same reasoning is worth nothing, so any debate gain must come from changing the reasoning.

4.3 Debate is not null — it is bidirectional

Mean accuracy over the nine cells is 0.496 for debate against 0.501 for the baseline: on the headline number, debate does nothing. That headline is misleading. Per cell (Table 1) debate moves accuracy from −0.100 to +0.095, and five of nine moves

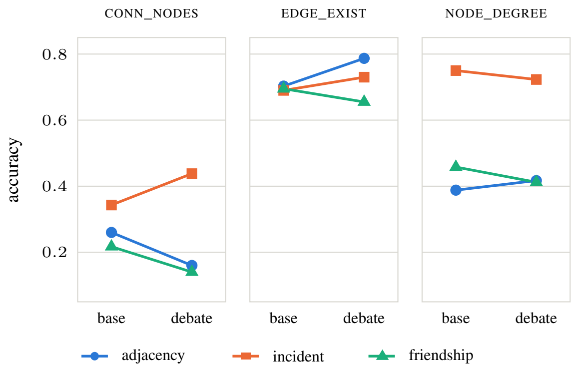

Figure 1: Debate fans the encodings apart: each line is one encoding's accuracy at baseline and under debate. On CONNECTED_NODES and EDGE_EXISTENCE the lines diverge — the best encoding gains, the others lose.

are significant — three losses, two gains. The near-zero mean is cancellation, not inertness.

The direction of that cancellation is the result. Debate *amplifies* encoding fragility (Table 2, Figure 1): on CONNECTED_NODES the max–min spread goes from 0.127 to 0.298, because debate lifts incident by +0.095 while dropping adjacency by −0.100 and friendship by −0.077. The gains land in the encoding that was already best.

The sharpest form of this is EDGE_EXISTENCE. At baseline it is encoding-insensitive ($Q = 0.52$, $p = 0.77$). Under debate the same task becomes significantly encoding-sensitive ($Q = 31.5$, $p = 1.4\text{e}{-07}$), with the spread growing from 0.013 to 0.132. Debate does not merely fail to remove fragility; on a task that had none, it manufactures it.

4.4 The amplification is chain-of-thought, not debate

The baseline answers directly while the debate Proposer writes a claim trace, so "debate vs. baseline" conflates CoT with debate. Turn 1 of every transcript *is* a single-turn CoT answer at identical decoding, which separates the two with no extra runs (Table 1).

The CoT step carries the movement, in both directions: +0.077 on EDGE_EXISTENCE/incident against −0.118, −0.090 and −0.052 on three others (four of nine significant, three of them losses). On the mean it is a net *loss* of 0.019 — the claim-trace scaffold is not free.

The loop step is smaller and better behaved: significant in two of nine cells, both positive (+0.060 and +0.055), worth +0.014 on the mean, and it never reverses the sign the scaffold set. The loop

Task	Encoding	Accuracy			CoT step		Loop step		Debate − base	
		base	turn 1	debate	Δ	p	Δ	p	Δ	p
CONNECTED_NODES	adjacency	0.260	0.142	0.160	−0.118	6e-08***	+0.018	0.145	−0.100	6e-07***
	incident	0.343	0.378	0.438	+0.035	0.231	+0.060	6e-06***	+0.095	4e-04***
	friendship	0.217	0.127	0.140	−0.090	2e-05***	+0.013	0.216	−0.077	2e-04***
EDGE_EXISTENCE	adjacency	0.703	0.732	0.787	+0.028	0.284	+0.055	0.009**	+0.083	3e-05***
	incident	0.690	0.767	0.730	+0.077	0.004**	−0.037	0.093	+0.040	0.079
	friendship	0.695	0.653	0.655	−0.042	0.073	+0.002	1.000	−0.040	0.039*
NODE_DEGREE	adjacency	0.388	0.402	0.417	+0.013	0.612	+0.015	0.223	+0.028	0.247
	incident	0.750	0.732	0.723	−0.018	0.410	−0.008	0.180	−0.027	0.215
	friendship	0.458	0.407	0.412	−0.052	0.041*	+0.005	0.775	−0.047	0.059
mean		**0.501**	0.482	**0.496**	−0.019		+0.014		−0.005	

Table 1: Per-cell accuracy, pooled over three dataset seeds ($n = 600$ per cell). *turn 1* is the Proposer's first answer — a single-turn chain-of-thought arm at identical decoding — so the *CoT step* (turn 1 − baseline) and *loop step* (final − turn 1) decompose *debate − base*. All p are paired McNemar ($^*p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$). Debate's mean effect is -0.005, but its per-cell effects run from -0.100 to $+0.095$ and five of nine are significant.

Task	std	max−min	Q	p
Baseline				
CONNECTED_NODES	0.053	0.127	65.8	5e-15
EDGE_EXISTENCE	0.006	0.013	0.5	0.77 NS
NODE_DEGREE	0.157	0.362	200.2	3e-44
Debate				
CONNECTED_NODES	**0.136**	**0.298**	190.1	5e-42
EDGE_EXISTENCE	**0.054**	**0.132**	31.5	1e-07
NODE_DEGREE	0.146	0.312	174.7	1e-38

Table 2: Cross-encoding spread per task; Q is Cochran's Q (df = 2), the paired omnibus test. Debate widens the spread on two of three tasks and turns EDGE_EXISTENCE from encoding-insensitive ($p = 0.77$) into significantly encoding-sensitive ($p =$ 1e−07).

is therefore not literally inert — it recovers part of what the scaffold costs — but it is a weak correction on top of a large format effect, and it does not touch the spread. Fragility amplification is a property of the *reasoning format*, not of the verification procedure layered over it.

The scaffold fails at declining to enumerate. Splitting CONNECTED_NODES by whether the gold answer is empty localizes the entire CoT penalty. On the 207 empty-gold instances the answer-only baseline is essentially perfect (0.990) while the claim-trace Proposer scores **0.077**; on the remaining 1,593 the scaffold is a net *gain* (0.180 to 0.234). Having written claims that name nodes, the model harvests those names into its answer — one instance states "There is no edge involving Robert" and answers "James, John, Michael." Fragility amplification survives the split: on the non-empty bucket the spread still widens, 0.141 to 0.258.

4.5 The verification asymmetry is absent

Debate's premise is that the Critic checks more reliably than the Proposer solves. We test it by cross-tabulating every verdict against whether the Proposer answer *that verdict judged* was correct (Table 3).

Pooled over 6,424 verdicts the Critic raises REVISE on 57.3% of correct answers and catches 59.6% of wrong ones. Against a base rate of 0.535 wrong, a REVISE moves $P(\text{wrong})$ only to 0.545: $\chi^2 = 3.36$, $p = 0.067$, $\phi = +0.023$, odds ratio 1.10. The verdict is worth about one accuracy point of information and does not clear significance even at N in the thousands. Three observations sharpen this beyond "the critic is weak."

The failure is calibration, not ignorance. On the three EDGE_EXISTENCE cells the Critic detects 95–96% of wrong answers — real discrimination, ϕ up to $+0.27$ — but simultaneously fires on 70–82% of correct ones. It knows something and cannot express it as a usable decision.

In two cells it is worse than chance. On NODE_DEGREE/adjacency the verdict is significantly *anti*-correlated with error ($\phi = -0.076$, $p = 0.048$, OR $= 0.73$), and CONNECTED_NODES/adjacency trends the same way. A loop acting on those verdicts is actively misinformed.

Its evidence is frequently fabricated. Told that the graph text is the only source of truth and to quote the offending edge, the Critic cites pairs that do not exist in the graph 4–26% of the time, worst on EDGE_EXISTENCE/adjacency. On the EDGE_EXISTENCE cells it usually cites no pair at all and writes prose (551 of 643 REVISEs on inci-

Task	Encoding	verdicts	Verdict quality: false alarm	Verdict quality: detection	ϕ	p	no pair cited	Cited evidence: not a real edge
CONNECTED_NODES	adjacency	741	0.664	0.586	-0.058	0.115	27	0.21
	incident	689	0.302	0.464	$+0.161$	3e-05	76	0.20
	friendship	711	0.517	0.547	$+0.019$	0.608	122	0.16
EDGE_EXISTENCE	adjacency	761	0.705	0.947	$+0.266$	2e-13	264	0.26
	incident	757	0.807	0.948	$+0.181$	7e-07	551	0.04
	friendship	790	0.816	0.965	$+0.221$	6e-10	484	0.07
NODE_DEGREE	adjacency	681	0.489	0.412	-0.076	0.048	5	0.10
	incident	615	0.138	0.218	$+0.098$	0.015	36	0.19
	friendship	679	0.463	0.496	$+0.033$	0.394	41	0.14
pooled		**6,424**	**0.573**	**0.596**	$+0.023$	**0.067** NS		

Table 3: Critic verdicts vs. whether the Proposer answer *that verdict judged* was correct. *False alarm* $= P(\text{REVISE} \mid \text{correct})$, *detection* $= P(\text{REVISE} \mid \text{wrong})$; ϕ, p from a 2×2 χ^2. *Not a real edge* is the fraction of cited pairs absent from the graph.

dent). The one sub-task the design assumed was easy — is a pair in a list — is where grounding fails most.

Downstream, the Proposer changes its answer after only 25–44% of REVISEs, and the net effect of every revision in the study is $+74$ corrections over 3,459 critiques.

4.6 How much headroom is left in the loop?

One might object that the loop is fine and only its stopping rule is wrong. We bound that with an oracle: per instance, the best answer any Proposer turn produced — perfect foresight over the turns that occurred. Pooled accuracy rises from 0.496 to 0.557 ($+0.062$), and that ceiling is distributed exactly the wrong way: $+0.105$ to $+0.150$ on the three EDGE_EXISTENCE cells (the non-fragile task) against $+0.012$ on NODE_DEGREE/incident, leaving CONNECTED_NODES spanning 0.160–0.467. No stopping rule, not even a clairvoyant one, recovers the encoding gap here.

5 Discussion

Debate's premise did not survive contact with the setting we chose *because* that setting was maximally favorable to it. The Critic's job here reduces to EDGE_EXISTENCE, a string lookup, on the task we independently measured to be easiest and the only encoding-insensitive one. It still could not turn that into a verdict better than noise, and it fabricated the evidence it was told to quote. This is stronger than the existing nulls: Huang et al. (2024) and Choi et al. (2025) show debate fails where verification is hard, and one may always answer that a checkable domain would differ. We built the checkable domain.

The second finding we did not expect. Interventions that raised mean accuracy raised the encoding spread with it: the CoT scaffold is worth up to $+0.077$ in the encoding that affords enumeration and costs up to 0.118 in ones that do not, and the oracle ceiling redistributes the same way. If this generalizes, robustness to serialization is not something better inference-time reasoning delivers for free — it may be in tension with it, because a scaffold is itself a format that some encodings suit better than others.

A methodological note that cost most of the project's effort: our first Proposer prompt contained a bug that presented as a finding. Its answer hint requested “a comma-separated list of node ids” for every encoding, but friendship labels nodes with names; the model complied, emitted integers, and 64 of 600 answers became unparseable — reading as evidence that friendship is hard to reason over, when it was a label-space mismatch the baseline never had. A later revision meant to strengthen all three roles instead drove the Proposer's capitulation rate from 0.63 to 0.96, costing 0.046 mean accuracy. Negative results about prompt-mediated methods are only as trustworthy as the prompt forensics behind them, and these failures are invisible in accuracy tables.

6 Conclusion

In a domain built so that atomic verification is trivial, a Proposer–Critic debate produced verdicts statistically independent of correctness, evidence fabricated up to a quarter of the time, no mean gain over greedy, and double the encoding spread on one task. The mechanism debate assumes is absent at this scale; what movement exists comes from the chain-of-thought scaffold. The next experiment is asymmetric — a 3B Proposer with a much larger Critic. If verification is still near chance there, the premise is in trouble.

Limitations

Scale. Every result is for a single 3B model in both roles, forced by an 11 GB VRAM ceiling. Verification may become cheap relative to generation only above some capability threshold: our result is that it is not cheap at 3B, not that it is never cheap. The asymmetric-critic experiment is the direct test, and we did not run it.

Critic granularity. Our Critic is holistic — it re-derives the answer and returns one verdict. Per-claim verification, one binary question per atomic claim aggregated in code, was not run, and the EDGE_EXISTENCE cells suggest it might behave differently: the Critic discriminates there (detection 0.95, $\phi = +0.27$) and only fails to turn that into a verdict.

Comparison budget. Majority vote was run on one of three seeds ($n = 200$ per cell), a weaker claim than the pooled $n = 600$ used elsewhere. Debate and majority vote are compared transitively through the shared baseline rather than head to head; since debate spends less than the vote (2.9 vs. 10 responses) and neither beats the baseline, no budget-matching scheme changes the conclusion.

Scope. One model family, one generator (ER), one size range, three encodings. FRIENDSHIP confounds non-integer labels with a social cover story that may invite transitive closure; an alphabetic-label encoding with neutral phrasing would separate them. Finally, the debate arm is one frozen prompt: given that a prompt bug once masqueraded as a finding (§5), we treat per-cell deltas as properties of *this* wording and the critic-quality measurement — which does not depend on the answer format — as the more portable result.

References

Hyeong Kyu Choi, Xiaoling Zhu, and Sharon Li. 2025. Debate or vote: Which yields better decisions in multi-agent large language models? In *Advances in Neural Information Processing Systems (NeurIPS)*.

Yilun Du, Shuang Li, Antonio Torralba, Joshua B. Tenenbaum, and Igor Mordatch. 2023. Improving factuality and reasoning in language models through multiagent debate. *arXiv preprint arXiv:2305.14325*.

Bahare Fatemi, Jonathan Halcrow, and Bryan Perozzi. 2024. Talk like a graph: Encoding graphs for large language models. In *The Twelfth International Conference on Learning Representations (ICLR)*.

Aric A. Hagberg, Daniel A. Schult, and Pieter J. Swart. 2008. Exploring network structure, dynamics, and function using NetworkX. In *Proceedings of the 7th Python in Science Conference (SciPy)*, pages 11–15.

Jie Huang, Xinyun Chen, Swaroop Mishra, Huaixiu Steven Zheng, Adams Wei Yu, Xinying Song, and Denny Zhou. 2024. Large language models cannot self-correct reasoning yet. In *The Twelfth International Conference on Learning Representations (ICLR)*.

Geoffrey Irving, Paul Christiano, and Dario Amodei. 2018. AI safety via debate. *arXiv preprint arXiv:1805.00899*.

Bryan Perozzi, Bahare Fatemi, Dustin Zelle, Anton Tsitsulin, Mehran Kazemi, Rami Al-Rfou, and Jonathan Halcrow. 2024. Let your graph do the talking: Encoding structured data for LLMs. *arXiv preprint arXiv:2402.05862*.

Qwen Team. 2024. Qwen2.5 technical report. *arXiv preprint arXiv:2412.15115*.

Xuezhi Wang, Jason Wei, Dale Schuurmans, Quoc V. Le, Ed H. Chi, Sharan Narang, Aakanksha Chowdhery, and Denny Zhou. 2023. Self-consistency improves chain of thought reasoning in language models. In *The Eleventh International Conference on Learning Representations (ICLR)*.

Jason Wei, Xuezhi Wang, Dale Schuurmans, Maarten Bosma, Brian Ichter, Fei Xia, Ed H. Chi, Quoc V. Le, and Denny Zhou. 2022. Chain-of-thought prompting elicits reasoning in large language models. In *Advances in Neural Information Processing Systems (NeurIPS)*.

A Prompt texts

Baseline (zero-shot). A per-task terse-format instruction, then GraphQA's question block verbatim (encoded graph, `Q: ...`, `A:`). For example, NODE_DEGREE uses *"Answer with a single integer (the degree) and nothing else."*

Proposer. *"Answer the question below using the graph. Build the answer as a numbered list of atomic claims — each claim one verifiable fact about a single edge (that it exists, or that no further edge involves the node) — then give the final answer. Number your claims 1., 2., 3., and so on, one claim per line. Then write a final line that begins with* `ANSWER:` *followed by .... Write nothing after that line."* The answer-format tail is deliberately label-space free ("written exactly as the graph writes them") so one wording covers all three encodings.

Critic. The Critic is told the graph text is the only source of truth, asked to independently enumerate the relevant edges from the list before comparing, to check in both directions (missed edges and invented edges), and to answer in exactly `VERDICT: AGREE` or `VERDICT: REVISE` followed by problems, each quoting an edge from the graph's edge list.

B Reproducibility

All numbers regenerate on CPU from committed run outputs. Raw model output is persisted per attempt, with full debate transcripts in trace sidecars, so scoring changes are re-applied without re-running the GPU.